

ADENTAN MUNICIPAL EDUCATION OFFICE

GovPay Desk

Individual contribution

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CSIT622 Capstone Project · Group 7

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Afrakumah Dapaah: Audit trail, documents and deployment

I own the audit trail, document handling, and the deployment.

The audit trail is append-only: every state change writes who acted, what they did, and when, and nothing in the application can edit or delete those rows. That is what allows the office to answer an auditor's question months later. Supporting documents are stored outside the public directory and served only through a controller that checks authorization first, so an unguessable URL is never the thing protecting an invoice.

Deployment was where the most instructive failures happened. A container that reports itself healthy while the proxy returns 502 gives you almost nothing to work with, so I made the failures loud: a startup banner naming what broke, an extension check that fails the build rather than warning on every request, and a health endpoint that answers without touching the database, so a database fault and a routing fault produce different symptoms.

The code this covers

- `app/Services/AuditLogger.php`
- `app/Http/Controllers/DocumentController.php`
- `Dockerfile`
- `docker/entrypoint.sh`
- `docs/DEPLOYMENT.md`

Questions to be ready for

The examiner may ask about any part of the system, not only this one. These are the ones closest to the work described above:

- What makes the audit trail trustworthy? What stops someone editing it?
- Why are uploaded documents not served directly from the public directory?
- Describe a deployment failure you diagnosed and how you found the cause.

Before submitting, confirm this describes what you actually did. This section is assessed individually, and you may be asked to demonstrate any claim in it. Rewrite anything that does not match your own work, and add what is missing, particularly what you found difficult and what you would do differently now.